Vol 2 Chapter 11 — Beyond the Standard Model: The Five Absences

Keeper (author pass)

2026-05-23 Saturday

Chapter 11 — Beyond the Standard Model: The Five Absences

Most theoretical-physics programs beyond the Standard Model have been about adding things. Grand unified theories added higher symmetries. Supersymmetry added partner particles for every Standard Model species. Various extensions added sterile neutrinos, magnetic monopoles, dark sectors, axions, hidden gauge groups. Each of these additions has been a target of major experimental searches, and most of them have produced negative results to date.

BST takes the opposite stance. The substrate's structure *forbids* these additions. The framework's Five-Absence Prediction Set identifies five major beyond-Standard-Model phenomena that BST structurally predicts do not exist:

1. No grand unified theory.
2. No proton decay.
3. No magnetic monopoles.
4. No sterile neutrinos.
5. No supersymmetric partner particles.

Each is independently falsifiable. Casey's Option 1 canonical 5-absence framework (Friday May 22, 2026), ratified at K65, makes this the framework's most distinctive predictive set.

11.1 No grand unified theory

The substrate's gauge structure $SU(3) \times SU(2) \times U(1)$ does not embed in any larger simple group at any energy scale. The structural reason: the substrate's BST primary integers (rank, N_c, n_C, C_2, g) are mathematically independent at their forcing arguments. A GUT embedding would require an additional integer to mediate the unification; the substrate has none.

Experimental confirmation: GUT-predicted phenomena (rapid proton decay, GUT-scale heavy bosons) have not been observed; current bounds are consistent with the substrate's no-GUT prediction.

11.2 No proton decay

Baryon number is a structural conservation law in BST (Volume 0 Chapter 8): the substrate K-type structure carries an exact global $U(1)_B$ symmetry that the substrate Hamiltonian commutes with universally. Proton decay would violate baryon number; baryon number is exact; therefore no proton decay.

Current experimental lower bound on the proton lifetime: $\tau_p > 10^{34}$ years. BST's structural prediction: the lifetime is *infinite* (proton is structurally stable). The two are consistent at any finite experimental precision.

11.3 No magnetic monopoles

Magnetic monopoles would correspond to topological excitations of the substrate's gauge sector. The substrate's gauge bundle on D_{IV}^5 is structurally trivial at the relevant level for monopole formation; the substrate's $SO(2)$ phase structure does not admit isolated magnetic charges.

Cabrera 1982 and many subsequent searches have produced no positive monopole detections. BST predicts they never will.

11.4 No sterile neutrinos

The substrate's neutrino structure is exhausted by the three left-handed neutrinos in the rank-2 weak doublets (Chapter 10). No right-handed (sterile) neutrino K-type exists in the substrate spectrum.

Recent experimental searches (LSND anomaly, MiniBooNE, IceCube) have been ambiguous. BST predicts no sterile neutrinos at any mass scale.

11.5 No supersymmetry

Supersymmetry would require partner particles paired by a transformation exchanging fermions and bosons. The substrate's K-type structure does not admit such a pairing; the spin-statistics distinction is structurally fixed by the Pin(2) $\mathbb{Z}_2$ grading (Volume 1 Chapter 4) and cannot be reorganized into supersymmetric pairings.

LHC searches for supersymmetric partners have produced negative results at the TeV scale. BST predicts no SUSY at any energy.

11.6 K65 ratification and the Casey Option 1 canonical

The five-absence framework was originally proposed with six absences (the original list included "no DM particle" alongside the others). Casey's Option 1 reformulation in May 2026 reorganized: dark matter is now a *positive prediction* (Volume 4 Chapter 10's $(3n_C + 1)/N_c = 16/3$ ratio derivation, with dark matter as substrate-natural rather than particle-physics-new), reducing the absences to a canonical five. K65 audit ratified this reformulation Friday May 22, 2026.

The K65 ratification anchors the substrate's K65-RATIFIED 4+1 scope: four primary absences (GUT, proton decay, monopoles, sterile neutrinos) plus one secondary (SUSY) per the substrate's structural derivation.

11.7 Five-Absence as falsifier package

Each absence is independently falsifiable. A single positive detection of any one phenomenon would refute the substrate-derivation at the relevant mechanism level:

- GUT discovery refutes substrate gauge-integer irreducibility.
- Proton decay refutes substrate K-type baryon-number conservation.
- Magnetic monopole detection refutes substrate gauge-bundle triviality.
- Sterile neutrino discovery refutes substrate K-type neutrino exhaustion.
- SUSY partner discovery refutes substrate Pin(2) $\mathbb{Z}_2$ grading.

The framework's experimental program tracks all five. Current experimental bounds are consistent with all five non-observations.

11.8 What comes next

Chapter 12 — Experimental Program — develops the substrate-engineering falsifier program (SP-29 Casimir, SP-30 substrate-engineering experiments) that targets BST's positive predictions for laboratory verification.

Where to look this up: Five-Absence Predictions Set: `notes/Five_Absence_Predictions_Principle.md`. K65 ratification: audit chain. Casey Option 1 canonical: Friday May 22, 2026 reframe. For standard BSM treatments: Particle Data Group reviews; Martin's *A Supersymmetry Primer* (hep-ph/9709356) for SUSY background.